

OUTDOOR

Outer surface
(Robin BC)

WALL

Inner surface
(Robin BC)

INDOOR

$$Q_{\text{solar}}(t) = \alpha_s G_s(t)$$

Outdoor air, $T_{\text{out}}(t)$

$$h_{\text{out}}(T_{\text{out}}(t) - T)$$

$$-k_{\text{eff}} \frac{\partial T}{\partial x} \Big|_{x=0} = h_{\text{out}}(T_{\text{out}} - T) + Q_{\text{solar}} \rightarrow$$

$$\rho c_p \frac{\partial T}{\partial t} = k_{\text{eff}} \frac{\partial^2 T}{\partial x^2}$$

$$T(x, t)$$

$$k_{\text{eff}} = k_{\text{dry}} + \alpha w_0$$

$$\rho, \quad c_p$$

x

$$T(x, 0) = T_{\text{mean}}^*(x)$$

Indoor air, T_{in}

$$h_{\text{in}}(T - T_{\text{in}})$$

$$\leftarrow -k_{\text{eff}} \frac{\partial T}{\partial x} \Big|_{x=L} = h_{\text{in}}(T - T_{\text{in}})$$

$x = 0$ Wall thickness L $x = L$